

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 10)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

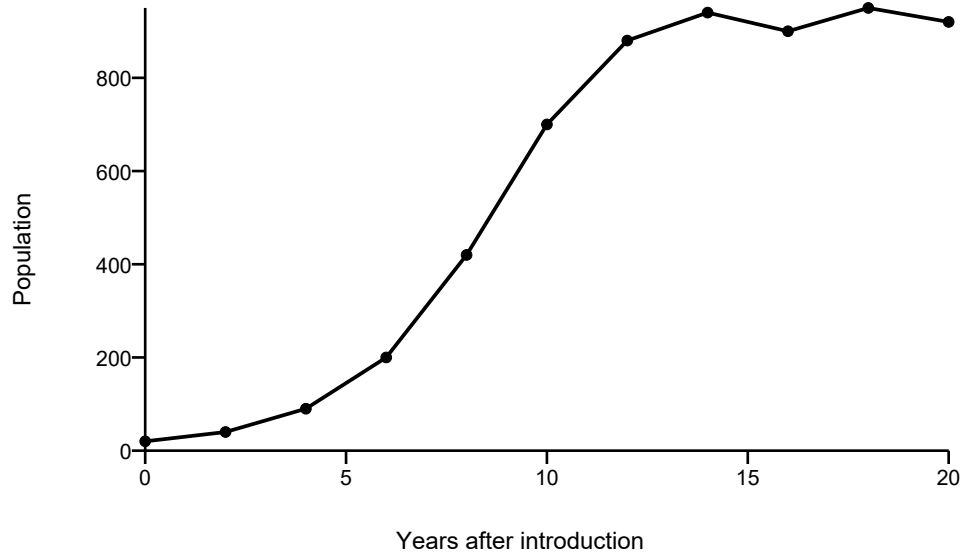
Question	Mark
1	
2	
3	
4	
5	
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7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Figure 1 shows the population of red deer on an island after they were introduced in year 0. No predators are present.

Figure 1



0 1

1

Name the shape of the growth curve and explain the changes in growth rate shown.

[4 marks]

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0 1

2

Explain the difference between density-dependent and density-independent factors, with an example of each.

[2 marks]

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0 1 . 3

Explain why the deer population on the island may need to be culled even though it appears stable.

[2 marks]

0 1 . 4

Explain how predator and prey populations regulate each other.

[2 marks]

10

0 2 . 1

Describe the ecological features of mangroves and the adaptations of mangrove trees.

[3 marks]

0 2 . 2

Explain **four** reasons why mangroves are important.

[4 marks]

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2

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4

0 2 . 3 Describe the main threats to mangroves and the conservation efforts used.
[3 marks]

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10

0 3 . 1 Describe a polyculture system using several species of carp and explain how it increases total productivity.
[4 marks]

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03 . 2

Explain how integrated multi-trophic aquaculture (IMTA) and aquaponics reduce pollution while increasing food production.

[4 marks]

03 . 3

Explain why tuna aquaculture cannot yet replace tuna fishing.

[2 marks]

10

04

Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha ⁻¹ yr ⁻¹	Food energy output / GJ ha ⁻¹ yr ⁻¹
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

04 . 1

Calculate the energy ratio (output / input) for each system.

[2 marks]

Answer

0 4 . 2

Use your answers to explain the difference between productivity and efficiency.

[3 marks]

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0 4 . 3

Explain what is meant by an **energy subsidy** and give **three** examples.

[3 marks]

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0 4 . 4

Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

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10

0 5 . 1

Explain how living organisms provide each of the following ecosystem services.

[6 marks]

Regulation of atmospheric composition

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Soil maintenance

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Interspecies relationships

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0 5 . 2

Explain why Darwin's orchid of Madagascar depends on a single moth species, and why such specialisation is both an advantage and a risk.

[2 marks]

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0 5 . 3

Explain why elephants are particularly important seed dispersers.

[2 marks]

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10

06 . 1

Complete Table 1 to show the environmental information that can be collected by satellite sensors detecting each type of radiation.

[3 marks]

Table 1

Radiation detected	Environmental information collected	
Reflected visible light		
Infrared emitted by the Earth's surface		
Reflected radar microwaves emitted by the satellite		

06 . 2

Explain why radar surveys are not interrupted by cloudy weather.

[1 mark]

06 . 3

Explain **two** advantages of satellite surveys over ground surveys for monitoring deforestation.

[2 marks]

06 . 4

Describe how motion-sensitive cameras and image databases could be used together to estimate the size of a tiger population.

[4 marks]

10

07 . 1

Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

1

2

07 . 2

Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

07 . 3

Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

5

08 . 1

Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

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08 . 2

Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

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08 . 3

Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

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10

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0	9
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1

Complete Table 1.

[6 marks]

Table 1

Method	Target species	One environmental disadvantage	
Demersal trawling			
Purse seining			
Pelagic long-lining			

0	9
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2

Explain why by-catch is a particular problem with shrimp trawling and drift netting.

[2 marks]

0	9	.	3
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Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

10

1 0 . 1

Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

Do not write outside the box

Extra space

10 . **2** State what is meant by **industrial symbiosis**.

[1 mark]

Do not write outside the box

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

OR

1 1 . 2

Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

END OF QUESTIONS